

Rational Matrix Policy Optimization for Rational Feed-Forward Language Models

Anonymous authors
Paper under double-blind review

Abstract

1 Introduction

2 Background and Related Work

3 Rational Matrix Policy Optimization

Let $\mathcal{L}(\theta)$ be the language-model training objective. For layer l , the RLB block projects the residual-stream input $x_l \in \mathbb{R}^d$ into G groups of width m ,

$$z_l = A_l x_l, \quad z_{l,g} = \text{group}_g(z_l), \quad (1)$$

normalizes each group by its RMS radius,

$$r_{l,g} = \sqrt{m^{-1} \|z_{l,g}\|_2^2 + \epsilon_{\text{rms}}}, \quad u_{l,g} = z_{l,g} / r_{l,g}, \quad (2)$$

and applies a learned rational map in normalized coordinates,

$$h_{l,g} = r_{l,g} R_{l,g}(u_{l,g}), \quad y_l = B_l \text{concat}_{g=1}^G h_{l,g}. \quad (3)$$

The input matrix A_l selects rational domains, the coefficients of $R_{l,g}$ shape the group response, and the output matrix B_l recombines rational features. MatrixPolicy uses this separation by partitioning parameters as

$$\theta = \theta_{\text{base}} \cup \theta_{\text{coeff}} \cup \{A_l, B_l\}_{l=1}^L. \quad (4)$$

The base parameters use the same optimizer as the non-rational baseline. Rational coefficients use a coefficient rule. MatrixPolicy is applied only to A_l and B_l .

In the homogeneous case $\epsilon_{\text{rms}} = 0$, and for nonzero group radii, each group has a positive scale gauge:

$$A_{l,g} \mapsto a_g A_{l,g}, \quad B_{l,g} \mapsto a_g^{-1} B_{l,g}, \quad a_g > 0. \quad (5)$$

The represented function is unchanged because $z_{l,g}$ and $r_{l,g}$ scale together, $u_{l,g}$ is unchanged, and the inverse scale in B_l cancels the scale in $h_{l,g}$. With a nonzero RMS floor, the exact quotient dynamics are replaced by the residual model in Appendix B. This matters because optimizer state is stored in coordinates, not in the quotient space of represented functions.

At step t , each matrix role $r \in \{\text{in}, \text{out}\}$ receives a role-depth factor $\rho_r(l)$ and an on-policy statistic factor $s_{\text{stat}}(l, r, t)$. The matrix update is

$$\begin{aligned} M_{l,r}^{t+1} = & M_{l,r}^t + \Delta_{\text{adapt}}(M_{l,r}; \eta_t a_{\text{mat}}(l, r, t)[1 - \mu(l, r, t)]) \\ & + \Delta_{\text{mat}}(M_{l,r}; \eta_t a_{\text{matnorm}}\mu(l, r, t)), \end{aligned} \quad (6)$$

where $M_{l,\text{in}} = A_l$, $M_{l,\text{out}} = B_l$, η_t is the shared base learning rate, Δ_{adapt} is an Adam-style adaptive step, and Δ_{mat} is a short early matrix-normalized step. The mixture coefficient

$\mu(l, r, t)$ is active only early in training. After the matrix update, MatrixPolicy computes a target log-ratio for each group and applies the bounded rebalance

$$A_{l,g} \leftarrow e^{\ell_g} A_{l,g}, \quad B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}. \quad (7)$$

The update changes the representative used by the optimizer while preserving, or with $\epsilon_{\text{rms}} > 0$ approximately preserving, the represented block function. Full hyperparameter definitions are in Appendix A.

Algorithm 1: Rational Matrix Policy step. Given gradients $\nabla \mathcal{L}(\theta^t)$ and base learning rate η_t :

1. Update exponential moving averages of rational-group pressures.
2. Compute role-depth and group-stat multipliers for each RLB matrix.
3. Update non-rational parameters with the base optimizer and update rational coefficients with the coefficient rule.
4. Update each A_l and B_l with the MatrixPolicy mixture step.
5. Apply the bounded positive-gauge rebalance to each rational group.

4 Experiments

5 Discussion

6 Conclusion

A MatrixPolicy Details

Role-depth factors. For layers indexed by $l = 1, \dots, L$ with $L > 1$, define normalized layer depth $d_l = (l - 1)/(L - 1)$ and set

$$\rho_{\text{in}}(l) = \text{clip}(1 - 0.50(d_l - 0.5), 0.55, 1.40), \quad (8)$$

$$\rho_{\text{out}}(l) = \text{clip}(1 + 1.00(d_l - 0.5), 0.55, 1.40), \quad (9)$$

$$a_{\text{role}}(l, r) = \max(0.10, 1 + 1.20(\rho_r(l) - 1)). \quad (10)$$

Let $\xi_t = t/T$ be training progress and let $\sigma_{a,b}(\xi)$ denote the smoothstep gate from 0 to 1 over $[a, b]$. The statistic multiplier $a_{\text{stat}}(l, r, t) > 0$ is the layer/role aggregate of the group preconditioners defined below, normalized to one when group statistics are not active. The local adaptive scale is

$$a_{\text{mat}}(l, r, t) = \text{clip}(3.0 a_{\text{role}}(l, r) a_{\text{stat}}(l, r, t), 0.40, 4.0). \quad (11)$$

The matrix-normalized branch uses multiplier $a_{\text{matnorm}} > 0$ and mixture weight $\mu(l, r, t) \in [0, 1]$, a smoothstep-gated function of ξ_t , layer depth, matrix role, and $a_{\text{stat}}(l, r, t)$.

On-policy group statistics. All ratios and logarithms below are evaluated on strictly positive stabilized RMS, EMA, and gain quantities; the displayed formulas omit the corresponding epsilon floors. For group g , MatrixPolicy tracks relative gradient pressures

$$p_{\text{in},g} = \frac{\text{rms}(\nabla A_{l,g})}{\text{rms}(A_{l,g})}, \quad p_{\text{out},g} = \frac{\text{rms}(\nabla B_{l,g})}{\text{rms}(B_{l,g})}, \quad p_{\text{rat},g} = \text{rms}(\nabla \theta_{\text{coeff},g}). \quad (12)$$

With EMAs $q_{\text{in},g}$, $q_{\text{out},g}$, and $q_{\text{rat},g}$, we define

$$\pi_g = \log q_{\text{in},g} - \log q_{\text{out},g}, \quad \alpha_g = \log q_{\text{rat},g} - \frac{1}{2}(\log q_{\text{in},g} + \log q_{\text{out},g}). \quad (13)$$

Group-stat preconditioning. The group preconditioner is centered and clipped:

$$c_g = c_{\text{gain},g} c_{\text{pressure},g} c_{\text{activity},g}, \quad c_g \leftarrow c_g / \text{geomean}(c), \quad c_g \leftarrow \text{clip}(c_g, 0.75, 1.35). \quad (14)$$

Here $k_g > 0$ is the role-specific group gain used for matrix preconditioning: the derivative-RMS gain for A_l and the output-RMS gain for B_l . The factors are

$$c_{\text{gain},g} = \left(\frac{\text{geomean}(k)}{k_g} \right)^{0.20}, \quad c_{\text{pressure},g} = \exp(0.10 d_{\text{pressure},g}), \quad (15)$$

where $d_{\text{pressure},g} = -\pi_g$ for A_l and $+\pi_g$ for B_l . The activity damping term is

$$c_{\text{activity},g} = \exp[-0.20 \text{ReLU}((\alpha_g - 0.05)/0.45)]. \quad (16)$$

Gauge rebalance. Let

$$\gamma_g = \log \text{rms}(A_{l,g}) - \log \text{rms}(B_{l,g}). \quad (17)$$

The target log-ratio is

$$\tau_g = \frac{1}{2}(\log d_g^{\text{gain}} - \log o_g^{\text{gain}}) + \beta_{\text{pressure}}(\log q_{\text{in},g} - \log q_{\text{out},g}). \quad (18)$$

Here $d_g^{\text{gain}} > 0$ and $o_g^{\text{gain}} > 0$ are the derivative and output group-gain estimates, and β_{pressure} is the pressure balance coefficient. Let $s(t, l) \in [0, 1]$ be the rebalance schedule, $a_g^{\text{activity}} \in (0, 1]$ the activity damping multiplier, and $\ell_{\text{max}} > 0$ the maximum absolute log-scale move. The exact target step is $\ell_g^* = \frac{1}{2}(\tau_g - \gamma_g)$, while the applied bounded move is

$$\ell_g = \text{clip}(s(t, l) a_g^{\text{activity}} \ell_g^*, -\ell_{\text{max}}, \ell_{\text{max}}). \quad (19)$$

For true RMS values, or when stabilization floors are inactive, the clipped update maps γ_g to $\gamma_g + 2\ell_g$; with active RMS floors this identity is approximate and the discrepancy is included in the residual perturbation model. The update is therefore a bounded move toward τ_g rather than exact canonicalization unless clipping and schedule damping are inactive. The transform $A_{l,g} \leftarrow e^{\ell_g} A_{l,g}$ and $B_{l,g} \leftarrow e^{-\ell_g} B_{l,g}$ preserves the homogeneous RLB function exactly under the nonzero-radius condition; the stabilized implementation is covered by the residual perturbation model in Appendix B.

B A Conditional Quotient-Optimization Separation

B.1 Quotient Geometry

For one RLB layer, suppress the layer index and write

$$z_g = A_g x, \quad (20)$$

$$r_g = \sqrt{m^{-1} \|z_g\|_2^2 + \epsilon_{\text{rms}}}, \quad (21)$$

$$u_g = z_g / r_g, \quad (22)$$

$$h_g = r_g R_g(u_g), \quad (23)$$

$$y = B \text{concat}_{g=1}^G h_g. \quad (24)$$

For $a_g > 0$, define

$$T_a(A_g, B_g) = (a_g A_g, a_g^{-1} B_g), \quad g = 1, \dots, G. \quad (25)$$

Equivalently, with $s_g = \log a_g$,

$$T_s(A_g, B_g) = (e^{s_g} A_g, e^{-s_g} B_g). \quad (26)$$

Let $\mathcal{P}$ be the local RLB-matrix parameter space, let $\mathcal{G} = (\mathbb{R}, +)^G$, and let $\mathcal{Q} = \mathcal{P}/\mathcal{G}$.

Lemma B.1 (Exact positive gauge). If $\epsilon_{\text{rms}} = 0$ and $r_g > 0$ for every group on the considered input, then $T_s(A, B)$ and (A, B) represent the same layer map for all $s \in \mathbb{R}^G$.

Proof. For one group, $A'_g = e^{s_g} A_g$ gives $z'_g = e^{s_g} z_g$. Since $\epsilon_{\text{rms}} = 0$, $r'_g = e^{s_g} r_g$, hence $u'_g = u_g$ and $h'_g = e^{s_g} h_g$. The corresponding output block satisfies $B'_g h'_g = e^{-s_g} B_g e^{s_g} h_g = B_g h_g$. Summing over groups gives the claim. $\square$

Assumption B.1 (Local quotient chart). In a neighborhood of the trajectory, parameters can be written as

$$\theta = \theta(q, s), \quad q \in \mathbb{R}^d, \quad s \in \mathbb{R}^G, \quad (27)$$

where q is a local coordinate on $\mathcal{Q}$ and s is the log-gauge coordinate. The objective satisfies

$$L(\theta(q, s)) = F(q). \quad (28)$$

For the exact gauge statements, $\epsilon_{\text{rms}} = 0$. Nonzero RMS stabilization is included only through the residual model of Assumption B.3.

Lemma B.2 (Gauge tangent orthogonality). Under Lemma B.1, for any differentiable loss depending on the RLB matrices only through the represented function, with Frobenius inner products,

$$\langle \nabla_{A_g} L, A_g \rangle - \langle \nabla_{B_g} L, B_g \rangle = 0. \quad (29)$$

Proof. Differentiate $L(e^{s_g} A_g, e^{-s_g} B_g) = L(A_g, B_g)$ at $s_g = 0$. $\square$

B.2 Canonicalized Quotient Preconditioning

For group g , define

$$\gamma_g = \log \text{rms}(A_g) - \log \text{rms}(B_g). \quad (30)$$

The logarithms in this subsection are taken on true positive RMS quantities. Stabilized RMS floors preserve the lemma exactly only when inactive; otherwise the resulting discrepancy is part of the perturbation model in Assumption B.3. Given a target log-ratio τ_g , the unclipped rebalance is

$$\ell_g = \frac{1}{2}(\tau_g - \gamma_g), \quad A_g^+ = e^{\ell_g} A_g, \quad B_g^+ = e^{-\ell_g} B_g. \quad (31)$$

Lemma B.3 (RMS-ratio canonicalization). For true positive RMS values, the update in equation 31 satisfies

$$\log \text{rms}(A_g^+) - \log \text{rms}(B_g^+) = \tau_g. \quad (32)$$

Proof. By homogeneity of RMS under positive scaling,

$$\log \text{rms}(A_g^+) - \log \text{rms}(B_g^+) = \log \text{rms}(e^{\ell_g} A_g) - \log \text{rms}(e^{-\ell_g} B_g) \quad (33)$$

$$= \gamma_g + 2\ell_g = \tau_g. \quad (34)$$

$\square$

Lemma B.4 (Gauge-dependent raw quotient step). Let $w > 0$, $a = e^s \sqrt{w}$, $b = e^{-s} \sqrt{w}$, and $\varphi(w) = \frac{1}{2}(w - y)^2$. One Euclidean gradient-descent step in (a, b) with step size η gives

$$w^+ = w - 2\eta w \cosh(2s)(w - y) + O(\eta^2). \quad (35)$$

Proof. Since $\partial\varphi/\partial a = (w - y)b$ and $\partial\varphi/\partial b = (w - y)a$,

$$w^+ = (a - \eta(w - y)b)(b - \eta(w - y)a) \quad (36)$$

$$= w - \eta(w - y)(a^2 + b^2) + O(\eta^2). \quad (37)$$

Using $a^2 + b^2 = w(e^{2s} + e^{-2s}) = 2w \cosh(2s)$ gives equation 35. $\square$

Assumption B.2 (Frozen quotient preconditioners). During the local phase, MatrixPolicy and AdamW induce fixed positive definite quotient preconditioners P_{MP} and P_{AdamW} satisfying

$$q_{t+1}^{\text{MP}} = q_t^{\text{MP}} - \eta_{\text{MP}} P_{\text{MP}} \nabla F(q_t^{\text{MP}}), \quad (38)$$

$$q_{t+1}^{\text{AdamW}} = q_t^{\text{AdamW}} - \eta_{\text{AdamW}} P_{\text{AdamW}} \nabla F(q_t^{\text{AdamW}}). \quad (39)$$

Here P_{MP} is the quotient preconditioner after gauge rebalance and P_{AdamW} is the quotient preconditioner induced by AdamW's raw-coordinate adaptive state.

B.3 Contraction and Perturbation Bounds

For $P \succ 0$, define the transformed quotient objective

$$\tilde{F}_P(x) = F(q_* + P^{1/2}x). \quad (40)$$

A preconditioned quotient step $q^+ = q - \eta P \nabla F(q)$ is equivalent to ordinary gradient descent on $\tilde{F}_P$ in the coordinate $x = P^{-1/2}(q - q_*)$.

Lemma B.5 (One-step transformed PL decrease). Let $P \succ 0$, define $\tilde{F}_P$ by equation 40, and set $x = P^{-1/2}(q - q_*)$ and $x^+ = x - L_P^{-1} \nabla \tilde{F}_P(x)$. Suppose the pathwise descent inequality

$$\tilde{F}_P(x^+) \leq \tilde{F}_P(x) - \frac{1}{2L_P} \left\| \nabla \tilde{F}_P(x) \right\|_2^2 \quad (41)$$

and the pointwise PL inequality

$$\frac{1}{2} \left\| \nabla \tilde{F}_P(x) \right\|_2^2 \geq \mu_P (\tilde{F}_P(x) - F_*), \quad 0 < \mu_P \leq L_P, \quad (42)$$

hold at the current iterate. If

$$q^+ = q - L_P^{-1} P \nabla F(q), \quad (43)$$

then

$$F(q^+) - F_* \leq \left(1 - \frac{\mu_P}{L_P}\right) (F(q) - F_*). \quad (44)$$

The descent condition equation 41 is implied by L_P -smoothness of $\tilde{F}_P$ on the segment from x to x^+ , but global smoothness or convexity is not required.

Proof. The update in q is the gradient step $x^+ = x - L_P^{-1} \nabla \tilde{F}_P(x)$. Combining equation 41 and equation 42 gives

$$\tilde{F}_P(x^+) - F_* \leq \left(1 - \frac{\mu_P}{L_P}\right) (\tilde{F}_P(x) - F_*). \quad (45)$$

Since $\tilde{F}_P(x) = F(q)$ and $\tilde{F}_P(x^+) = F(q^+)$, this proves equation 44. $\square$

Proposition B.1 (Time-varying PL quotient bound). For each optimizer $o \in \{\text{MP}, \text{AdamW}\}$, suppose both ideal quotient trajectories start from the same quotient point q_0 and have the form

$$q_{t+1}^o = q_t^o - L_{o,t}^{-1} P_{o,t} \nabla F(q_t^o), \quad P_{o,t} \succ 0, \quad (46)$$

and that Lemma B.5 applies at every step with constants $(L_{o,t}, \mu_{o,t})$. Define

$$\kappa_{o,t} = \frac{L_{o,t}}{\mu_{o,t}}, \quad \Xi_o(T) = \prod_{t=0}^{T-1} (1 - \kappa_{o,t}^{-1}). \quad (47)$$

Then

$$F(q_T^o) - F_* \leq \Xi_o(T) (F(q_0) - F_*). \quad (48)$$

Consequently, if $\Xi_{\text{MP}}(T) < \Xi_{\text{AdamW}}(T)$, MatrixPolicy has the smaller certified finite-horizon PL contraction bound.

Proof. Apply Lemma B.5 at each step and multiply the resulting one-step factors. $\square$

Corollary B.1 (Frozen PL quotient bound). Suppose Assumptions B.1 and B.2 hold. For $o \in \{\text{MP}, \text{AdamW}\}$, suppose Lemma B.5 applies along the transformed ideal trajectory with $P = P_o$ and fixed constants (L_o, μ_o) . Let

$$\kappa_o = \frac{L_o}{\mu_o}, \quad \chi_o = 1 - \kappa_o^{-1}. \quad (49)$$

If $\kappa_{\text{MP}} < \kappa_{\text{AdamW}}$ and $\eta_o = 1/L_o$, then

$$\chi_{\text{MP}} < \chi_{\text{AdamW}} < 1 \quad (50)$$

and, for every initial point whose two ideal trajectories remain in their local regions,

$$F(q_T^{\text{MP}}) - F_* \leq \chi_{\text{MP}}^T (F(q_0) - F_*), \quad (51)$$

$$F(q_T^{\text{AdamW}}) - F_* \leq \chi_{\text{AdamW}}^T (F(q_0) - F_*). \quad (52)$$

Proof. This is Proposition B.1 with constant $P_{o,t} = P_o$, $L_{o,t} = L_o$, and $\mu_{o,t} = \mu_o$. The map $\kappa \mapsto 1 - \kappa^{-1}$ is strictly increasing for $\kappa \geq 1$. $\square$

Lemma B.6 (Preconditioned quadratic contraction). Let

$$F(q) = F_\star + \frac{1}{2}(q - q_\star)^\top H(q - q_\star), \quad H \succ 0. \quad (53)$$

For $P \succ 0$, define

$$K(P) = P^{1/2}HP^{1/2}, \quad \kappa(P; H) = \frac{\lambda_{\max}(K(P))}{\lambda_{\min}(K(P))}, \quad \rho(P; H) = \frac{\kappa(P; H) - 1}{\kappa(P; H) + 1}. \quad (54)$$

With

$$\eta^\star(P) = \frac{2}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))}, \quad (55)$$

the update $q_{t+1} = q_t - \eta^\star(P)P\nabla F(q_t)$ satisfies

$$F(q_T) - F_\star \leq \rho(P; H)^{2T}(F(q_0) - F_\star), \quad (56)$$

and the bound is sharp.

Proof. Let $e_t = q_t - q_\star$ and $\tilde{e}_t = P^{-1/2}e_t$. Then

$$\tilde{e}_{t+1} = (I - \eta^\star(P)K(P))\tilde{e}_t, \quad 2(F(q_t) - F_\star) = \tilde{e}_t^\top K(P)\tilde{e}_t. \quad (57)$$

Diagonalize $K(P) = U\Lambda U^\top$ and set $v_t = U^\top \tilde{e}_t$. The largest multiplier satisfies

$$\max_i |1 - \eta^\star(P)\lambda_i| = \frac{\lambda_{\max}(K(P)) - \lambda_{\min}(K(P))}{\lambda_{\max}(K(P)) + \lambda_{\min}(K(P))} = \rho(P; H). \quad (58)$$

Thus

$$2(F(q_{t+1}) - F_\star) = \sum_i \lambda_i v_{t+1,i}^2 \quad (59)$$

$$\leq \rho(P; H)^2 \sum_i \lambda_i v_{t,i}^2 \quad (60)$$

$$= 2\rho(P; H)^2(F(q_t) - F_\star). \quad (61)$$

Iteration gives equation 56. Equality holds when the transformed initial error $\tilde{e}_0 = P^{-1/2}(q_0 - q_\star)$ is an eigenvector of $K(P)$ associated with $\lambda_{\min}(K(P))$ or $\lambda_{\max}(K(P))$. $\square$

Theorem B.1 (Conditional worst-case optimization separation). Suppose Assumptions B.1 and B.2 hold, and suppose F is the quadratic objective in Lemma B.6. Assume also that

$$\kappa(P_{\text{MP}}; H) < \kappa(P_{\text{AdamW}}; H), \quad (62)$$

and set $\eta_{\text{MP}} = \eta^\star(P_{\text{MP}})$ and $\eta_{\text{AdamW}} = \eta^\star(P_{\text{AdamW}})$. For every integer $T \geq 1$, let U_T be a sufficiently small quotient neighborhood of $q_\star$ on which both ideal trajectories remain in the local chart through time T . Then

$$\sup_{\substack{q_0 \in U_T \\ q_0 \neq q_\star}} \frac{F(q_T^{\text{MP}}) - F_\star}{F(q_0) - F_\star} = \rho(P_{\text{MP}}; H)^{2T} < \rho(P_{\text{AdamW}}; H)^{2T} = \sup_{\substack{q_0 \in U_T \\ q_0 \neq q_\star}} \frac{F(q_T^{\text{AdamW}}) - F_\star}{F(q_0) - F_\star}. \quad (63)$$

Proof. The map $\kappa \mapsto (\kappa - 1)/(\kappa + 1)$ is strictly increasing for $\kappa > 0$. The strict condition-number inequality therefore gives $\rho(P_{\text{MP}}; H) < \rho(P_{\text{AdamW}}; H)$. Lemma B.6 gives the sharp worst-case factors. Since the quadratic ratios are invariant under scaling $q_0 - q_\star$, the extremal transformed-eigenvector initializations can be chosen inside any sufficiently small neighborhood U_T . $\square$

Assumption B.3 (Uniform local perturbation bound). For a fixed horizon T , let the ideal quotient recursions be

$$q_{t+1}^o = \Psi_{o,t}(q_t^o), \quad o \in \{\text{MP}, \text{AdamW}\}. \quad (64)$$

The implemented quotient iterates satisfy

$$q_{t+1}^{o,\text{impl}} = \Psi_{o,t}(q_t^{o,\text{impl}}) + r_t^o, \quad \|r_t^{\text{MP}}\|_2 + \|r_t^{\text{AdamW}}\|_2 \leq \delta \quad (65)$$

for all $0 \leq t < T$. The residuals include the RMS floor, denominator floors, clipping, moment lag, weight decay, and nonquadratic Taylor remainders.

Proposition B.2 (Stability of a strict finite-time gap). Fix $T \geq 1$ and suppose the idealized dynamics satisfy

$$m_T := F(q_T^{\text{AdamW}}) - F(q_T^{\text{MP}}) > 0. \quad (66)$$

Assume the implemented and ideal recursions use the same initial points, and assume the ideal trajectories lie in a compact set $K \subset \text{int}(U)$, where U is a compact neighborhood on which F and all $\Psi_{o,t}$ are Lipschitz. Then there exists $\delta_0 > 0$ such that, whenever Assumption B.3 holds with $\delta < \delta_0$, the implemented trajectories remain in U through time T and

$$F(q_T^{\text{AdamW,impl}}) - F(q_T^{\text{MP,impl}}) > \frac{m_T}{2} > 0. \quad (67)$$

Proof. Let C_Ψ be a common Lipschitz constant on U for the finite family of ideal one-step maps $\{\Psi_{o,t} : o \in \{\text{MP}, \text{AdamW}\}, 0 \leq t < T\}$. Let $r_{\text{stay}} = \text{dist}(K, U^c) > 0$. By induction, as long as the implemented trajectories remain in U ,

$$\|q_t^{\text{MP,impl}} - q_t^{\text{MP}}\| + \|q_t^{\text{AdamW,impl}} - q_t^{\text{AdamW}}\| \leq C_T \delta \quad (68)$$

for a constant C_T depending only on T and C_Ψ . Taking $\delta \leq r_{\text{stay}}/(2C_T)$ keeps the implemented trajectories in U and closes the induction. If C_F is a Lipschitz constant for F on U , then

$$\left| F(q_T^{\text{MP,impl}}) - F(q_T^{\text{MP}}) \right| + \left| F(q_T^{\text{AdamW,impl}}) - F(q_T^{\text{AdamW}}) \right| \leq C_F C_T \delta. \quad (69)$$

Taking

$$\delta_0 = \min \left\{ \frac{m_T}{2C_F C_T}, \frac{r_{\text{stay}}}{2C_T} \right\} \quad (70)$$

gives the claim. $\square$